

Quantum-Topological Locking of the Space-Time Product

—Unified Relationship Among Space, Time, the Speed of Light and Angular Momentum

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Abstract

Within the Angular-Momentum Network Geometry (ANG-TOE) framework, this paper establishes a rigid identity connecting quantum effects, spacetime and gravitation:

Where:

- $S_g = h$ is the elementary quantum of angular momentum (Planck's constant).
- F denotes force as perceived by humans.
- $c(\Phi)$ is the local speed of light, determined by topological density $\Phi = |\text{Link}|/\chi$.
- $dx \cdot dt$ represents the product of space and time.

We demonstrate the following three propositions:

1. Ontological layer (6D angular-momentum manifold): No space, no time; only angular-momentum quanta S_g and linking networks exist.
2. Perceptual layer (4D spacetime projection): The product of space $\times$ time equals the ratio between the angular-momentum quantum (S_g) and force (F).

3. Topological-density modulation: The local speed of light $c(\Phi)$ varies with topological density, which in turn modulates the reading of the spacetime product.

This identity locks Planck's constant, the local speed of light, force, space and time into one geometric constraint, providing a new geometric foundation for understanding gravitational time dilation, dark matter and dark energy.

Keywords: spacetime product, angular-momentum quantum, local speed of light, topological density, ANG-TOE, gravitational time dilation

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1. Introduction: Spacetime Is Not Fundamental; Angular Momentum Is

Humans measure space in meters, time in seconds, and the speed of light in meters-per-second. Within ANG-TOE, however, all these quantities are projected readings rather than the ontological foundation of the universe.

At the underlying cosmic layer (Layer-0), there exists no space and no time, only the angular-momentum network together with one rigid constraint:

$$J_{\text{total}} \equiv 0$$

When the angular-momentum quantum ($S_g = \hbar$) projects from the 6-dimensional angular-momentum manifold onto the 4-dimensional human perceptual layer, it splits into four observable quantities: space, time, force and energy.

The core question of this paper: What does space $\times$ time correspond to in the projected layer? The answer is that it equals the ratio of the angular-momentum quantum to force, modulated by the local speed of light.

2. Angular-Momentum Quantum and the Song Unit

2.1 Definition

- Sg (Song-unit): The elementary quantum unit of angular momentum.
- Song: The underlying cosmic vacuum superfluid, which cannot be directly observed.
- Sg: Quantifies angular-momentum excitations generated within the Song medium.

2.2 Dimensionality

$$[\text{Sg}] = \text{Energy} \times \text{Time} = \text{Angular Momentum}$$

3. Human Perceptual Layer: Space, Time and Force

On the human perceptual layer (4D spacetime projection), the relationship between angular momentum, space, time and force is derived below.

3.1 Angular Momentum and Force

Under the approximation of constant force:

$$J = F \cdot r \cdot t$$

3.2 Angular Momentum, Space and Time

$$r \cdot t = \frac{J}{F}$$

3.3 Quantization

Since angular momentum is quantized as $J = n \cdot \text{Sg}$:

For a single angular-momentum quantum ($n = 1$):

4. Topological Modulation of the Local Speed of Light

4.1 The Speed of Light Is Not a Universal Constant

Within the ANG-TOE framework, the local speed of light $c(\Phi)$ is a function of topological density Φ :

Where:

- c_0 : Reference vacuum speed of light (constant).
- $\Phi = |\text{Link}|/\chi$: Local topological density (number of link windings / Euler characteristic).

4.2 Coupling Between Space and Time

Within the 4D projection, space and time are coupled via the local speed of light:

$$dr = c(\Phi) \cdot dt$$

Hence:

5. Unified Formula: Locking of the Space-Time Product

Combine the quantized spacetime product with local-light-speed coupling:

5.1 Full Causal Chain

5.2 Physical Implications

Parameter	Direction of Change	Effect on the spacetime product $r \cdot t$
Topological density Φ	Increase	$c(\Phi)$ decreases; modifies $r \cdot t$
Force F	Increase	$r \cdot t$ decreases
Angular-momentum quantum S_g	Fixed	Rigid geometric anchor point

6. Verification: Geometric Interpretation of Gravitational Time Dilation

In general relativity, clocks run slower within strong gravitational fields. ANG-TOE re-interprets this phenomenon geometrically:

- Near massive celestial bodies, the local topological density Φ increases.
- The local speed of light $c(\Phi)$ decreases.

3. To satisfy $r \cdot t = Sg/F$, if spatial distance r is fixed, time t must change accordingly.
4. This variation manifests as “time dilation” within the human perceptual frame.
- This effect does not originate from the curvature of spacetime; it is a coupled reading arising from the angular-momentum quantum and the scaled local speed of light.
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7. Implications for Dark Matter and Dark Energy

7.1 Dark Matter

Dark matter is the residual force reading produced by angular-momentum wakes projected onto space:

$$\rho_{\text{DM}} \propto \frac{Sg}{c(\Phi) \cdot r^2}$$

7.2 Dark Energy

Dark energy is the cosmic expansion reading generated by global residual cancellation of angular momentum projected onto the time dimension:

$$\rho_{\Lambda} \propto \frac{Sg}{c(\Phi) \cdot t^2}$$

8. Falsifiable Predictions

Prediction	Falsification Criterion	Test Method
The local speed of light $c(\Phi)$ varies with topological density Φ	$c(\Phi)$ is independent of Φ	High-precision atomic-clock comparison
The identity $r \cdot t = Sg/F$ holds across all scales	Deviation greater than 1%	Precise mechanics-time coupling experiments
Dark-matter density is proportional to $1/r^2$	Observed deviation exceeds 3σ	Galaxy rotation-curve measurements
Gravitational time dilation arises from light-speed scaling, not spacetime curvature	Non-GR spacetime patterns are detected	Gravitational-wave observation

9. Conclusion

This formula links the quantum (S_g), geometry (Φ), dynamics (F) and spacetime (r, t) within a single equation.

The spacetime product is not an independent physical quantity — it is the expanded reading of the angular-momentum quantum inside the projection layer.   

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End of Paper